Pipeline Evaluation Framework

1. Purpose

This framework defines how to evaluate the project pipeline from input papers to knowledge graph, CMOCs, cross-study synthesis, and final programme theory.

It directly answers the professor's main concern:

How do we know the difference between the LLM output and the human Richmond outcome, and what metrics will we use at each step?

2. Evaluation Principle

The project is not evaluated only at the final report stage.

Each pipeline step must have:

1. A defined input.
2. A defined output.
3. A gold or reference standard.
4. A metric.
5. A human review point.
6. A refinement action if the output is wrong.

This makes the project auditable. If the final output is wrong, we can identify whether the failure came from corpus coverage, extraction, graph construction, CMOC formation, synthesis, or reporting.

3. Final Target of the Project

The final target is not simply to "build a knowledge graph."

The final target is to reproduce and evaluate a machine-assisted realist synthesis pipeline against Richmond et al.'s human realist review. The pipeline should:

1. Represent the same 28-study corpus.
2. Extract clinically meaningful Context, Intervention, Mechanism, and Outcome concepts.
3. Build directed, typed, evidence-supported relationships among those concepts.
4. Generate CMOCs and assign them to Richmond-style programme theory statements.
5. Produce a final synthesis comparable to Richmond's human conclusion.
6. Show where the model agrees, disagrees, misses evidence, or over-generates claims.

The knowledge graph is therefore an intermediate evidence structure, not the final research contribution by itself.

4. Pipeline Stages and Evaluation Design

Stage	**Input**	**Output**	**Gold / Reference**	**Metrics**	**Human Review**
0. Review question and initial theory	Richmond topic and project question	Review question, scope, initial programme theory	Richmond aim and realist review method	Scope alignment; traceability to Richmond	Confirm project target and inclusion logic.
1. Search and screening	Candidate papers or benchmark registry	Included/ excluded study list	Richmond 28 included papers and PRISMA counts	Recall, precision, F1, false negatives	Review all exclusions and uncertain studies.
2. Document ingestion	20 PDFs plus 8 abstract/metadata records	Text corpus and document registry	`data/studies_metadata.jsonl`; 28-study inventory	Document coverage, extraction success, missing-text count	Confirm each study is represented.
3. GraphRAG indexing	Text corpus	Chunks, embeddings, graph extraction input	28-study corpus	Indexed document count; chunk coverage	Inspect missing or malformed documents.
4. Entity extraction	Text chunks and prompts	Raw KG nodes	E01-E47	Entity precision, recall, F1, category accuracy, evidence support	Map raw nodes to E-codes; reject unsupported nodes.
5. Relationship extraction	Raw nodes and text evidence	Directed typed edges	R01-R40	Strict edge F1, relaxed edge F1, direction accuracy, predicate accuracy, evidence support	Check source, target, predicate, direction, and evidence.
6. CMOC extraction	Papers, KG evidence, ontology	Per-paper CMOCs	Human-adjudicated CMOC gold; interim PTS1-PTS5 alignment	CMOC tuple accuracy, PTS assignment, polarity accuracy, quote support	Review study-level CMOCs and error labels.
7. Cross-study synthesis	Per-paper CMOCs	Demi-regularities, contradictions,	Richmond five-context programme	PTS coverage, chain fidelity, contradiction	Confirm synthesis does not flatten

Stage	Input	Output	Gold / Reference	Metrics	Human Review
		PTS synthesis	theory	handling, negative mechanism capture	context differences.
8. Final programme theory	Synthesized CMOCs and evidence	Final report / programme theory	Richmond final conclusion and discussion	Claim coverage, unsupported-claim rate, alignment with five contexts	Human signoff before final use.
9. Refinement loop	Metrics and human feedback	Revised prompts, mappings, outputs	Prior scored run	Delta in metrics; error reduction	Decide whether to rerun, accept, or stop.

5. Stage 0: Review Question and Initial Theory

Evaluation question:

Does the pipeline address the same realist question as Richmond?

Reference:

- Richmond's review asks how educational interventions develop analytical and non-analytical clinical reasoning, and why they work differently for different learners and contexts.
- Source: `data/paper-Richmond-original.pdf`, Abstract and Introduction.

Required output:

- Review question.
- Scope statement.
- Initial programme theory.
- Inclusion/exclusion criteria.

Metric:

Metric	Pass Rule
Scope alignment	Must include educational interventions, clinical reasoning, undergraduate medical/healthcare students, and analytical/non-analytical reasoning.
Realist framing	Must ask what works, for whom, why, and under what circumstances.
Traceability	Each scope decision must be linked to Richmond or professor-approved project scope.

6. Stage 1: Search and Screening

Evaluation question:

Can the pipeline recover the same included corpus as Richmond?

Gold:

- 28 included papers.
- Richmond reports 149 full texts retrieved and 28 ultimately included.
- Local source: `data/studies_metadata.jsonl`.

Current project status:

- `main.py` currently uses benchmark mode, which automatically includes all 28 Richmond papers.
- This is acceptable for downstream benchmarking, but it does not evaluate a real screening agent.

Metrics:

Metric	**Formula**
Screening recall	Richmond included papers recovered / 28.
Screening precision	Correct included papers / all papers included by the model.
False negatives	Richmond papers incorrectly excluded.
False positives	Non-Richmond papers incorrectly included.

Pass rule:

For the benchmark version, recall is expected to be 28/28 because the included set is provided. For a production search/screening version, false negatives require manual review.

7. Stage 2: Document Ingestion

Evaluation question:

Are all 28 Richmond studies available to the pipeline in a usable text form?

Reference:

- 20 local PDF full texts.
- 8 abstract/metadata-only records.
- Source: `outputs/evidence_base/study_inventory.tsv`.

Current project concern:

- `graphrag/input` contains 28 text records.

- `outputs/graphrag_data/stats.json` reports 20 indexed documents.
- This must be resolved before official KG evaluation.

Metrics:

Metric	**Pass Rule**
Study registry coverage	28/28 records present.
Full-text extraction coverage	20/20 PDFs extracted.
Abstract record coverage	8/8 abstract records represented and labelled.
Index coverage	Official KG index should represent 28/28 records.

Refinement action:

If fewer than 28 records are indexed, rebuild the KG index and produce a document-level coverage report before continuing.

8. Stage 3: GraphRAG Indexing

Evaluation question:

Does GraphRAG receive and index the correct corpus?

Required output:

- Document registry.
- Chunk table.
- Entity extraction input.
- Graph extraction logs or stats.

Metrics:

Metric	**Meaning**
Indexed document count	Number of documents represented in graph index.
Chunk coverage	Whether each study contributes chunks.
Empty document count	Documents with no usable extracted text.
Evidence traceability	Whether later nodes/edges can point back to document/chunk evidence.

Pass rule:

No official entity or relationship score is reported unless corpus coverage is first confirmed.

9. Stage 4: Entity Extraction

Evaluation question:

Does the model extract Richmond-relevant concepts, and can those concepts be mapped to E01-E47?

Gold:

- E01-E47 in `data/gold_standard.json`.

Required output:

- Raw node list.
- Node category.
- Evidence source.
- Mapping to E-code or rejection reason.

Metrics:

Metric	**Target**
Strict entity precision	Count only exact E-code matches.
Strict entity recall	Count recovered E-codes out of 47.
Relaxed entity F1	Allow approved synonyms and abstraction-level matches.
Category accuracy	Correct C/I/M/O family assignment.
Evidence support rate	Accepted entities should have source evidence.
Unmapped rate	Should be reviewed and reduced over refinement rounds.

Current baseline:

- `outputs/Phase1_Evaluation.md` reports E-code coverage of 0.66.
- This is useful as an interim coverage metric, but it is not yet a complete KG node verification score because raw GraphRAG nodes still need systematic E-code mapping.

Refinement action:

If entity recall is low, inspect missing E-codes by category and update prompts, ontology examples, or normalization rules.

10. Stage 5: Relationship Extraction

Evaluation question:

Does the model produce directed, typed, evidence-supported relationships that match Richmond's programme theory?

Gold:

- R01-R40 in `data/gold_standard.json`.

Required output:

```
source_e_code
predicate
target_e_code
source_document
evidence
confidence
```

Metrics:

Metric	Meaning
Strict edge precision	Exact source-predicate-target matches / predicted triples.
Strict edge recall	Gold R01-R40 triples recovered / 40.
Strict edge F1	Overall exact edge performance.
Direction accuracy	Correct direction among conceptually matched edges.
Predicate accuracy	Correct relation type among matched source-target pairs.
Relaxed edge F1	Allows approved predicate aliases and human-approved equivalent entities.
Evidence support rate	Edges with source evidence / accepted edges.

Current gaps:

- `outputs/relationships_long.json` uses local within-CMOC IDs, not E-code triples.
- `scripts/evaluate_phase1.py` does not yet compute strict edge fidelity against R01-R40.
- `core/ontology.py` must be reconciled with the gold predicate `CONSTRAINS`.

Refinement action:

Materialize relationships into E-code triples, score against R01-R40, then inspect missing edges and wrong-predicate edges separately.

11. Stage 6: CMOC Extraction

Evaluation question:

Can the model extract Context-Mechanism-Outcome configurations from each study in a way that supports Richmond's theory?

Gold:

- Interim: Richmond PTS1-PTS5 and E-code coverage.

- Strong: a new human-adjudicated per-paper CMOC gold table.

Required output:

```
paper_id
context
intervention
mechanism_resource
mechanism_response
outcome
pts_assignment
polarity
evidence
```

Metrics:

Metric	Meaning
CMOC count per paper	Detects under-extraction and over-extraction.
Tuple accuracy	Correct context, mechanism, outcome combination.
PTS assignment accuracy	Correct assignment to PTS1-PTS5.
Polarity accuracy	Correct positive, negative, mixed, or unclear label.
Quote/evidence support	Accepted CMOCs supported by paper text.
Multi-CMOC rate	Whether the model avoids one-CMOC-per-paper oversimplification.

Current status:

- Current outputs report either 56 or 57 CMOCs depending on artifact.
- `outputs/evidence_table.md` and `outputs/PRISMA_report.md` report 57 CMOCs.
- `outputs/Phase1_Evaluation.md` reports 56 CMOCs.
- This mismatch must be reconciled before using CMOC count as an official result.

Refinement action:

Create a single canonical CMOC output file, score it, and log differences between generated CMOCs and human-adjudicated CMOCs.

12. Stage 7: Cross-Study Synthesis

Evaluation question:

Does the model synthesize across studies in the same conceptual structure as Richmond?

Gold:

- Richmond's five-context programme theory.
- PTS1-PTS5 in `data/gold_standard.json`.

Required output:

- PTS-specific synthesis sections.
- Demi-regularities.
- Contradiction register.
- Evidence table linking claims to studies.

Metrics:

Metric	Pass Rule
PTS coverage	Must recover all 5 Richmond PTS categories.
Chain fidelity	CMOCs should follow valid realist chains.
Negative mechanism capture	Must include low confidence, cognitive load, stress, and negative outcomes where supported.
Contradiction handling	Must identify where the same intervention works differently under different contexts.
Evidence traceability	Each synthesis claim should link to supporting studies.

Current baseline:

- `outputs/Phase1_Evaluation.md` reports PTS coverage of 1.0.
- It also reports ontology chain fidelity of 0.411, which indicates that coverage alone is not enough. The model may mention all PTS categories while still producing weak or incomplete chains.

Refinement action:

Prioritize chain fidelity and negative-mechanism correctness, not only PTS coverage.

13. Stage 8: Final Programme Theory and Report

Evaluation question:

Does the final report match Richmond's human conclusion without making unsupported claims?

Gold:

- Richmond final conclusion and Discussion.
- Five context-specific programme theory statements.

Metrics:

Metric	Meaning
Final claim coverage	Does the report include Richmond's main conclusions?
Context sensitivity	Does it avoid one-size-fits-all claims?

Metric	Meaning
Analytical/non-analytical balance	Does it discuss both reasoning forms and when each is appropriate?
Unsupported claim rate	Claims without evidence / all claims.
Omission list	Richmond conclusions missing from model output.
Contradiction list	Model claims that conflict with Richmond.

Pass rule:

The final report must recover the Richmond conclusion that intervention effectiveness depends on learner knowledge, confidence, self-efficacy, coping, feedback, and opportunities to practice reasoning in authentic or simulated settings.

14. Stage 9: Human-in-the-Loop Refinement

Incorrect outputs are handled through a controlled refinement loop.

Recommended loop:

1. Run one pipeline stage.
2. Score the output.
3. Human reviewer labels errors.
4. Localize the error source.
5. Update only the relevant prompt, schema, mapping rule, or extraction logic.
6. Rerun the affected stage.
7. Compare metrics before and after.
8. Freeze the improved output or log why it remains unresolved.

Error localization table:

If the Error Is...	Likely Cause	Refinement Action
Missing study	Corpus/indexing problem	Rebuild document registry and GraphRAG index.
Missing entity category	Prompt or ontology example too weak	Add category examples and mapping rules.
Wrong E-code	Normalization problem	Improve E-code mapping table and human examples.
Wrong edge direction	Relation prompt/schema problem	Add directed triple examples and direction tests.
Wrong relation type	Predicate vocabulary problem	Freeze relation set and add alias mapping.
Missing negative mechanism	Model bias toward positive	Add negative CMOC examples

If the Error Is...	Likely Cause	Refinement Action
	findings	and polarity scoring.
Weak final theory	Cross-study synthesis problem	Require PTS-specific evidence and contradiction handling.
Unsupported claim	Reporting problem	Enforce source-linked claims only.

Current project gap:

The existing orchestrator has human-in-the-loop checkpoints, but benchmark behavior can auto-approve outputs. For final evaluation, auto-approval is replaced by explicit review records.

15. Minimum Publication-Ready Evaluation Package

Before project results are claimed, the following package is produced:

Artifact	Why It Matters
Corpus coverage report	Shows all 28 Richmond studies are represented.
Richmond gold memo	Defines what the human benchmark is.
Entity mapping table	Makes node verification auditable.
Relationship triple table	Makes edge verification auditable.
CMOC adjudication table	Creates per-paper extraction gold.
Pipeline metrics report	Shows performance at each stage.
Error analysis report	Shows what failed and how it was refined.
Final Richmond comparison memo	Shows agreement, disagreement, omissions, and added claims.

16. Immediate Next Work Plan

The next work should follow this order:

1. Freeze the Richmond programme-theory gold package: E01-E47, R01-R40, and PTS1-PTS5.
2. Reconcile the 56 vs 57 CMOC mismatch in current outputs.
3. Verify or rebuild GraphRAG indexing so all 28 records are represented.
4. Convert generated relationships into directed E-code triples.
5. Add strict and relaxed relationship metrics against R01-R40.
6. Create the per-paper CMOC adjudication table, starting with the 20 full-text PDFs and marking the 8 abstract-only records separately.
7. Run the full pipeline evaluation and produce a stage-by-stage metrics report.
8. Use human review to refine the weakest stage before writing final project claims.

17. Current Status Summary

What the project already has:

- A 28-study Richmond benchmark registry.
- 20 PDF full-text papers and 8 abstract/metadata records.
- A Richmond-derived gold entity set: E01-E47.
- A Richmond-derived gold relationship set: R01-R40.
- Five programme theory statements: PTS1-PTS5.
- A working multi-agent pipeline that generates CMOCs and reports.
- Preliminary Phase 1 metrics.

What the project still needs:

- Verified 28-document KG indexing.
- Strict entity and relationship verification tables.
- E-code materialization for relationship outputs.
- Reconciled CMOC counts.
- Human-adjudicated per-paper CMOC gold.
- Explicit human review records instead of auto-approval for official evaluation.

18. Conclusion

This framework turns the project from "we built a graph" into a defensible research workflow:

Richmond is the human reference. The KG is an intermediate structure. The final contribution is an evaluated pipeline that shows whether LLM-assisted realist synthesis can recover Richmond's programme theory, where it agrees, where it fails, and how human feedback improves it.

19. Citation and Evidence Policy

This framework uses three levels of evidence:

1. Richmond et al. (2020) as the human realist synthesis benchmark.
2. GraphRAG and related graph-based retrieval work as the technical rationale for graph-supported sensemaking.
3. Local repository artifacts as implementation evidence for what the current pipeline actually does.

The final project paper should distinguish local project evidence from external scholarly evidence. Local artifacts are reported as implementation evidence, while published papers justify the research design, evaluation logic, and methodological contribution.

20. References

Richmond, A., Cooper, N., Gay, S., Atiomo, W., & Patel, R. (2020). The student is key: A realist review of educational interventions to develop analytical and non-analytical clinical reasoning ability. Medical Education, 54(8), 709-719. https://doi.org/10.1111/medu.14137

Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., & Larson, J. (2025). From local to global: A GraphRAG approach to query-focused summarization. arXiv:2404.16130v2.

Professor project abstract. (2026). Proposed AI Agent Pipeline for Systematic Review in Education. Local file: `documents/abstract-from-professor/abstract.md`.

Professor verification plan. (2026). Plan for model verification using Richmond et al. as benchmark. Local file: `documents/plan_verification_text.txt`.